

DEFORMATION BEHAVIOUR OF ZIRCALOY-4 BETWEEN 77 AND 900 K

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Abstract—Tensile tests were carried out on zircaloy-4 over a range of temperatures 77–900 K, flow stress values at various strain rates were used to determine the activation parameters. Two thermally activated zones were observed between 77 and 600 K and the deformation continues to be thermally activated above 700 K. Within the temperature range 600–700 K, an athermal region occurs with a hump, the position of which is a function of strain rate. The various mechanisms which can be responsible for the thermal and athermal zones are discussed in the light of experimental results. The mechanical behaviour of zircaloy-4 and zirconium–oxygen alloys were observed to be generally similar.

Résumé—Nous avons étudié les variations de la contrainte d'écoulement du zircaloy-4 dans le domaine de température 77–900 K par essai de traction. Les paramètres d'activation des mécanismes contrôlant la déformation plastique ont été déterminés.

Dans le domaine 77–600 K, on observe deux zones de déformation thermiquement activée. Entre 600 et 700 K, on observe la superposition d'un plateau et d'un maximum de durcissement dû à un effet inverse de la vitesse de déformation. Au-dessus de 700 K la déformation est à nouveau thermiquement activée. Les différents mécanismes à l'origine des domaines thermiques et athermiques sont discutés à la lumière de la composition de l'alliage et des résultats expérimentaux. Le comportement mécanique du zircaloy-4 est, dans l'ensemble, assez proche de celui des alliages zirconium–oxygène.

Zusammenfassung—Die Fließspannung von Zirkaloy-4 wurde im Temperaturbereich von 77 bis 900 K im Zugversuch gemessen; die Aktivierungsparameter wurden bestimmt.

Im Temperaturbereich 77–600 K werden zwei Bereiche thermisch aktivierter Verformung beobachtet. Zwischen 600 und 700 K findet man die Überlagerung eines Plateaus mit einem Verfestigungsmaximum, welches umgekehrt von der Verformungsgeschwindigkeit abhängt. Oberhalb 700 K ist die Verformung wieder thermisch aktiviert. Die verschiedenen thermischen und athermischen Mechanismen, die für die verschiedenen Temperaturbereiche verantwortlich sein könnten, werden anhand der Zusammensetzung der Legierung und der experimentellen Ergebnisse diskutiert. Das mechanische Verhalten von Zirkaloy-4 gleicht insgesamt gesehen dem der Zirkon–Sauerstofflegierungen.

1. INTRODUCTION

The deformation behaviour of zircaloy-4 has been investigated mainly at high temperature, where a strong strain-rate dependence of the flow stress was observed and associated with a super plasticity phenomenon which takes place near the temperature of the two phases region ($\alpha + \beta$) $\approx 0.5 T_m$, where T_m is the melting temperature [1, 2]. But very little is known about the behaviour of zircaloy-4 at intermediate temperatures, in a region which includes the operating range of most of zircaloy-4 applications. On the other hand, a considerable amount of works has been carried out on the effect of interstitial solutes in pure zirconium on its mechanical behaviour for a wide range of temperatures. Most of these interstitial atoms are more or less present in zircaloy-4 in addition to essential elements such as Sn, Fe and Cr. The individual effects of these elements on pure zirconium will be summarized below.

Interstitial atoms in α -Zr are generally recognised as having a marked effect in increasing the flow stress

between low-temperatures and up to half the transformation ($\alpha + \beta$) temperature. The low temperature deformation (77–600 K) is known to be controlled by two thermally activated processes [3–8], at 0.5 at.% of interstitial the activation energy is about $\mu b^3/6$ and the activation volume $22b^3$ [9]. Soo and Higgins [3], Ruano and Elssner [6] and Das Gupta *et al.* [8] have reported that the deformation was thermally activated up to ambient where an athermal plateau was observed, then the deformation continues to be thermally controlled up to 600 K but several authors, such as Mills and Craig [10] (zirconium–oxygen), Sastry *et al.* [11] (α -zirconium) and Tyson *et al.* [12] (zirconium–nitrogen–oxygen), obtained different results. Indeed they report that only one thermally activated mechanism controls the deformation up to 600 K. The athermal region in the flow-stress against temperature occurring between 600 and 700 K with a hump was reported for polycrystalline α -zirconium oxygen alloys [13], for zircaloy-2 [12], for single-crystal α -zirconium oxygen alloys and for a zirconium–1 wt.% niobium alloy [15]. Furthermore, it was

found that a difference of interstitial content as small as 150 ppm was sufficient to change the flow stress.

The respective effects of nitrogen and oxygen solute atoms were shown to be similar [3, 10, 16]. Hydrogen has a different effect due to the fact that at room-temperature, a zirconium hydride precipitate forms which leaves a negligible amount of H in solid solution [7] and a precipitation hardening mechanism can be evoked.

The effect of tin as substitution element in the deformation behaviour of zirconium was examined by Luton and Jonas [17] in series of alloys containing up to 5% Sn at high temperature above 900 K. They show that the strengthening effect of tin could be due to an increase in the height of local obstacles which, in turn, can be associated with a reduction of stacking fault energy due to Sn addition. However, no strengthening effect could attributed to the increase in obstacles density which accompanies solute additions. Other additional elements exist in zircaloy-4 such as Fe and Cr, little is known about their effect on deformation behaviours at various temperatures. Nevertheless, it was indicated [18–19] that Fe could be responsible for the static strain ageing peak observed in zircaloy-2 and in zircaloy-4 at 725 K.

2. EXPERIMENTAL PROCEDURE

Cold rolled zircaloy-4 in the form of sheets of commercial purity was supplied by Ugine-Aciars (France). Its chemical analysis is shown in Table 1. Tensile specimens with a gauge length of 25 mm and a cross-section of $6 \times 1.15 \text{ mm}^2$ were machined from the sheets; the specimen length is in the rolling direction. The as-deformed material was annealed in secondary vacuum (5.10^{-6} – 10^{-5} torr) during 3 h at 925 K and a mean grain size of about $15 \mu\text{m}$ was then found. After pickling the specimens were tested using an Instron tensile machine.

The low temperature tests were carried out in the following baths: boiling liquid nitrogen (77 K), regulated pentane (148 K), regulated methylic alcohol

(178, 223, 248 K), melting ice (273 K), regulated water (298, 323 K) and boiling water (373 K).

The high temperature tests (473–873 K) were performed in the Instron high temperature vacuum capsule where a high purity argon was introduced after maintaining the primary vacuum.

Each specimen was held respectively 30 min and 10 min at the temperature test prior to deformation at high and low temperatures. The specimen temperature was controlled through a chromel–alumel thermocouple mounted at the head of the tensile specimen. Tensile tests at both constant and variable cross-head speeds were used: 0.05, 1 and 5 mm/min. The true stress-strain curves were computed. For each temperature several tests (from 2–5) were carried out and a dispersion from 1–3% on flow-stress values was found.

3. RESULTS

Figures 1(A) and 1(B) show the evolution of the stress-strain curves versus temperature. A yield drop appears from 270 K and an inverse temperature effect is observed between 573 and 673 K for a plastic strain lower than 0.04. Figures 2(A) and 3(A) show the evolution of the flow-stress (at 0.02 plastic strain against temperature for various strain rates). In Figs. 2(B) and 3(B), the variation of the shear stress τ is shown (a correction is introduced to account for the variation of the shear modulus μ with temperature).

The shear stress τ is obtained from the following relationship:

$$\tau = \sigma_{0.02}/M \quad (1)$$

Where $\sigma_{0.02}$ is the applied tensile stress and M the average Taylor's factor which relates the normal strain rate $\dot{\epsilon}$ to the shear strain rate $\dot{\gamma}$ in the glide plane. According to Luton and Jonas [20], M is assumed to be equal to 4.

At low and middle temperatures (below about 600 K) the flow stress shows a strong temperature dependence: in this range we observe a discontinuity in the linear temperature dependence of flow stress. Below about 180 K, the flow stress decreases sharply with increasing temperature whereas above this temperature it decreases more slowly.

In the range 600–700 K, the flow stress becomes almost independent of temperature and strain rate, then presents a maximum and decreases again as temperature rise proceeds. The maximum position is strain rate dependent. When the strain rate rises, the maximum moves towards high temperatures. This behaviour suggests that, between 600 and 700 K, the deformation rate-controlling mechanism is 'athermal', a thermally activated ageing process taking place while, above and below this region, the strong flow-stress dependence on temperature and strain rate is characteristic of thermally activated processes. Similar behaviour has been observed for zirconium–oxygen alloys (3–15).

Table 1. Chemical composition of zircaloy-4 (ppm by weight unless otherwise stated)

Sn = 1.38%
Fe = 0.20%
Cr = 0.10%
C = 110 ppm
O = 1445
Si = 40
Ta = < 100
Ni = < 50
Nb = < 50
Al = 44
W = 50
Pb = 20
N = 25

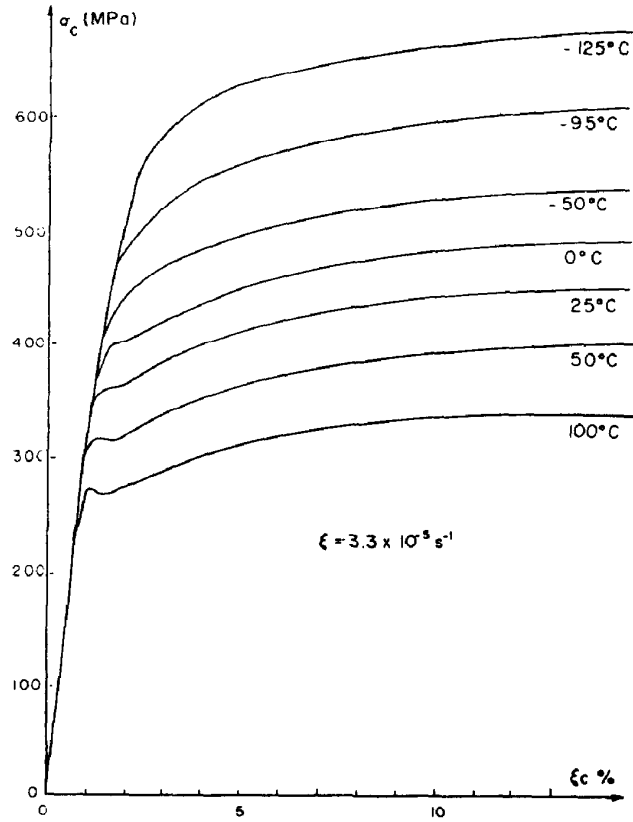


Fig. 1(A) Applied stress as a function of strain in the temperature range 77–373 K ($\dot{\epsilon} = 3.3 \times 10^{-5} \text{ s}^{-1}$).

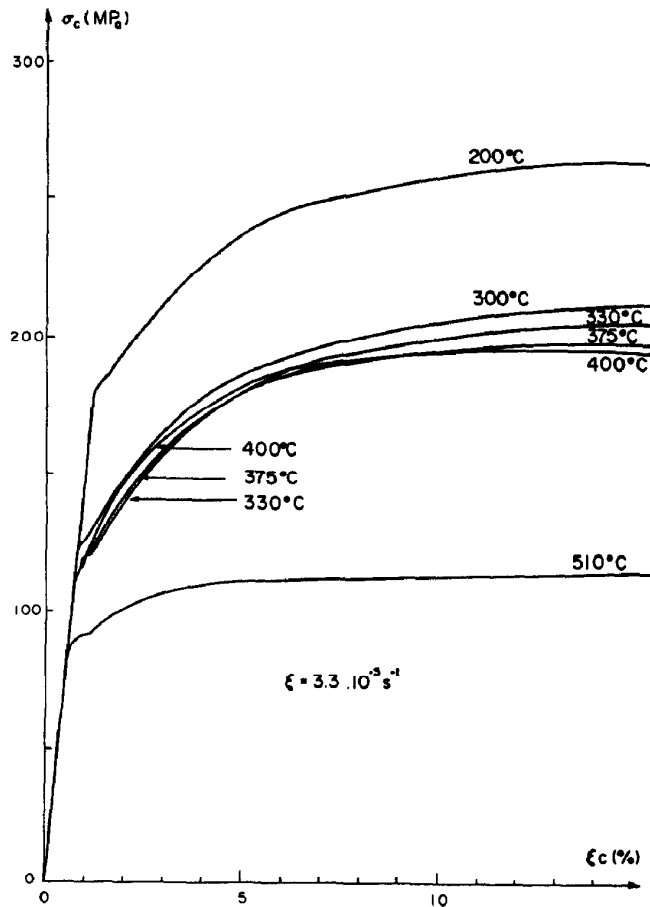


Fig. 1(B) Applied stress as a function of strain in the temperature range 373–783 K ($\dot{\epsilon} = 3.3 \times 10^{-5} \text{ s}^{-1}$).

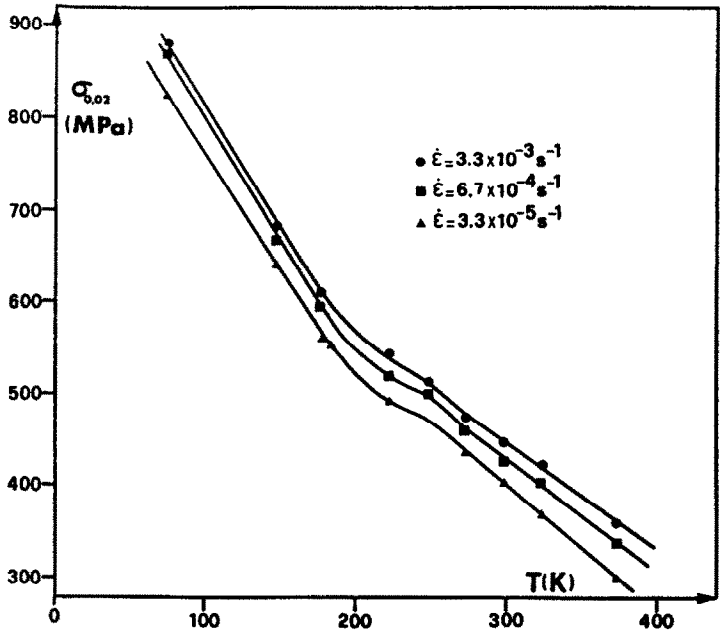


Fig. 2(A) Flow stress at 0.02 plastic strain as a function of temperature 77–373 K at various strain rates.

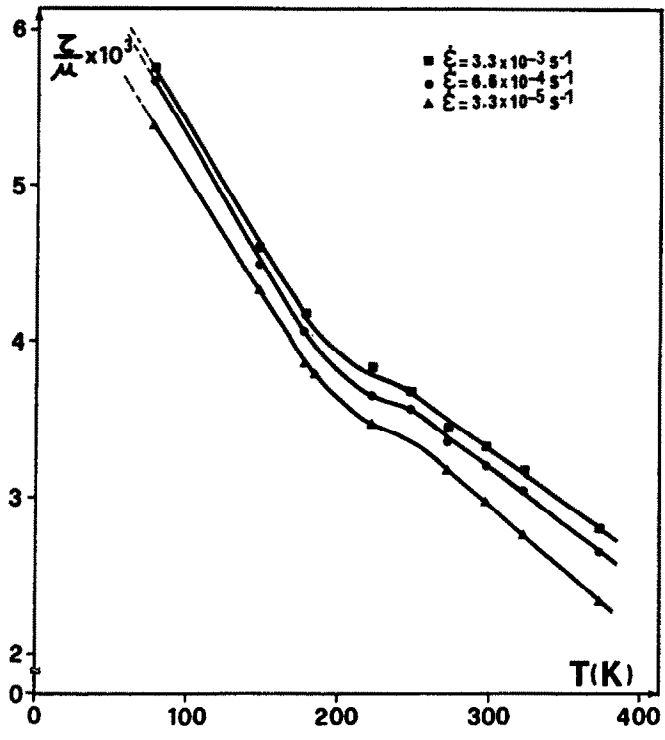


Fig. 2(B) The applied shear stress τ (corrected to the shear modulus variations with temperature) as a function of temperature (77–373 K) at various strain rates.

Calculation of thermodynamic parameters

In the present analysis τ was used to calculate the various thermodynamic parameters. The apparent activation volume associated with a deformation process has been obtained from the following classical equation [21]:

$$V_{app} = kT \left(\frac{\partial \ln \dot{\gamma}}{\partial \tau} \right)_T \simeq kT \left(\frac{\Delta \ln \dot{\epsilon}}{\Delta \tau} \right)_T \tag{2}$$

where k is Boltzman's constant, $\dot{\gamma}$ is the shear strain rate, τ is the applied shear stress and T is the absolute temperature.

In Schoeck's theoretical treatment of thermally activated plastic flow [22], it has been shown that in case of a temperature dependent local force acting on the dislocation segment at its pinned point, the apparent activation enthalpy for the process can be given

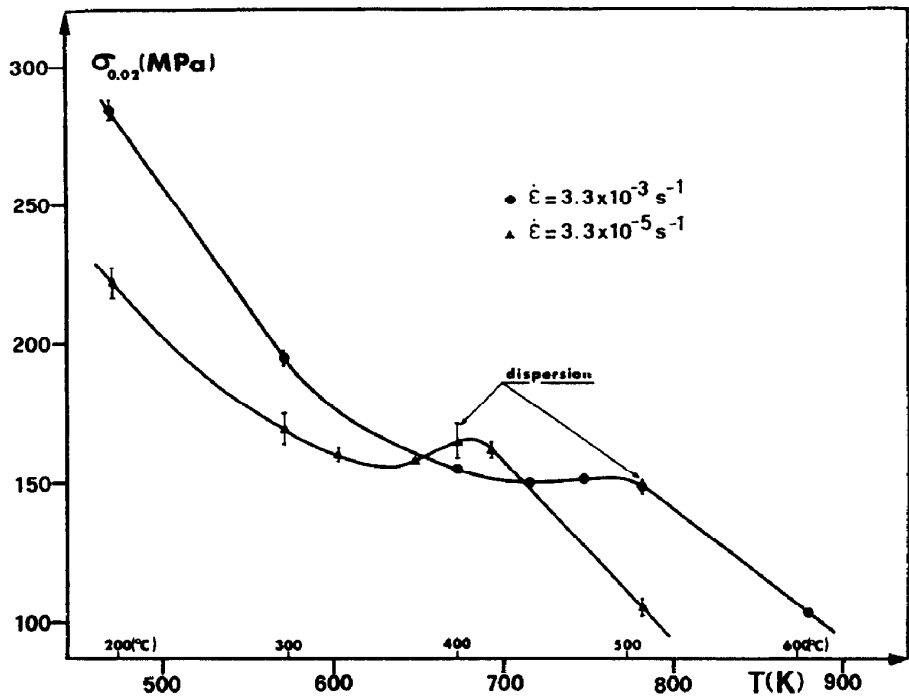


Fig. 3(A) Flow stress at 0.02 plastic strain as a function of temperature (473–880 K) at various strain rates.

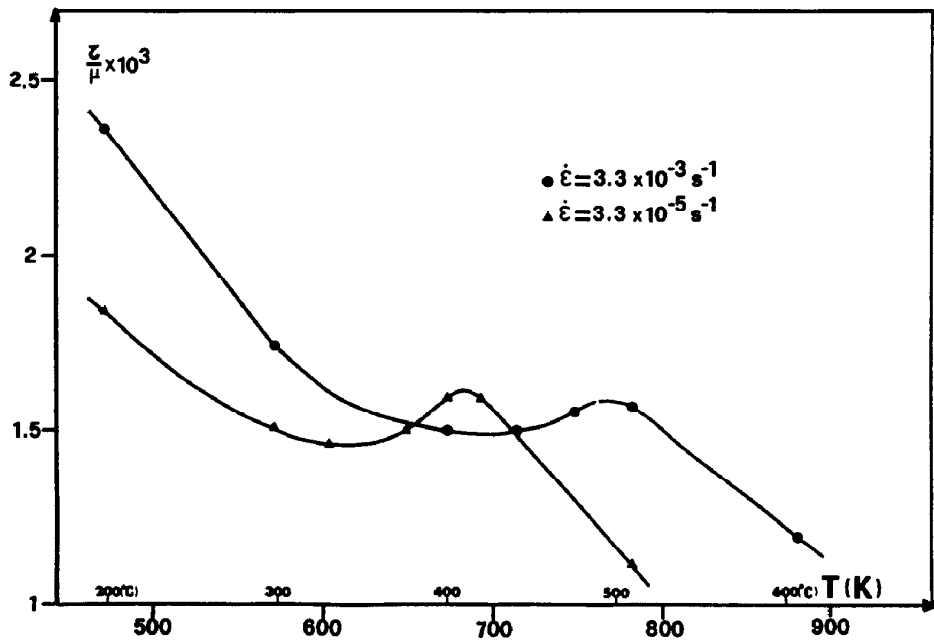


Fig. 3(B) The applied shear stress τ (corrected to the shear modulus variations with temperature) as a function of temperature (473–880 K) at various strain rates.

by:

$$\Delta H_{app} = -TV_{app}\left(\frac{\partial \tau}{\partial T}\right)_{\dot{\gamma}}$$

or

$$\Delta H_{app} = kT^2\left(\frac{\partial \ln \dot{\gamma}}{\partial T}\right)_\tau \simeq kT^2\left(\frac{\Delta \ln \dot{\epsilon}}{\Delta T}\right)_\tau$$

and the apparent activation free energy by:

$$(3) \quad \Delta G_{app} = \frac{\Delta H_{app} + \alpha V_{app} \tau}{1 - \alpha} \tag{5}$$

with

$$\alpha = \frac{T}{\mu} \left(\frac{\partial \mu}{\partial T} \right)$$

or

$$\Delta G_{app} = \frac{\mu}{\mu_0} \int_{\tau'}^{\tau_M} V_{app} d\tau' \tag{6}$$

with

$$\tau' = \frac{\mu_0}{\mu} \tau$$

Where μ is the shear modulus corresponding to the temperature of τ and μ_0 the shear modulus corresponding to the temperature of τ_M (taken at 77 K) equation (6) is used to determine ΔG_{app} for each value of τ and thus for each corresponding value of T .

Using the data of Figs. 2(A) and 3(A) and the strain rate changes (equation 2) to calculate the apparent activation volumes as a function of temperature and stress [Figs. 4(A) and 4(B)] it is found that V_{app} varies between 10 and 400 b^3 (both methods yield the same values). These values are in good agreement with experimental ones proposed for Zr-O [3, 5, 8 & 13], Zr-Nb [6], Zr-H [7] and α -Zr [11].

ΔH_{app} was computed using both equations (3) and (4) and plotted vs temperature in Fig. 5. At low and middle temperatures ($T < 400$ K), Equations (3) and (4) give same values as can be seen in Fig. 5, but for $T > 400$ K results given by equation (3) are greater than the ones obtained by equation (4).

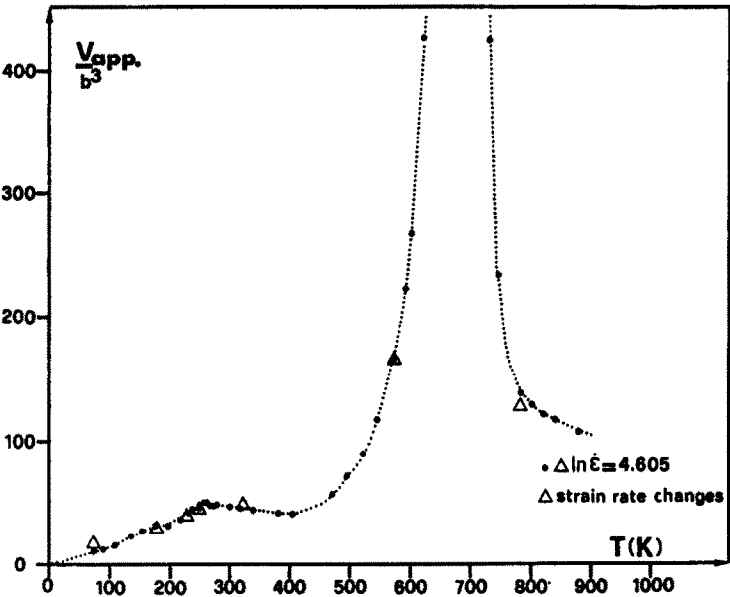


Fig. 4(A) The apparent activation volume as a function of temperature.

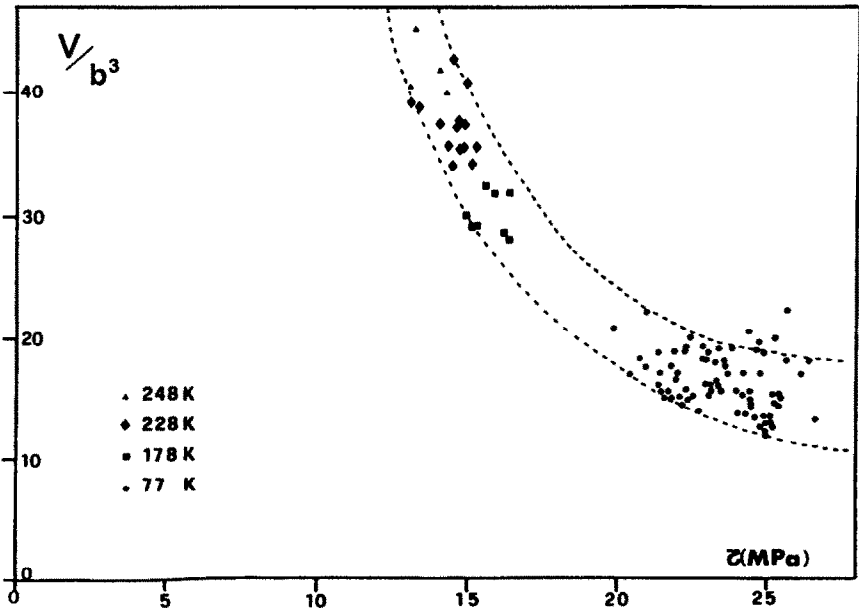


Fig. 4(B) The apparent activation volume as a function of applied shear stress.

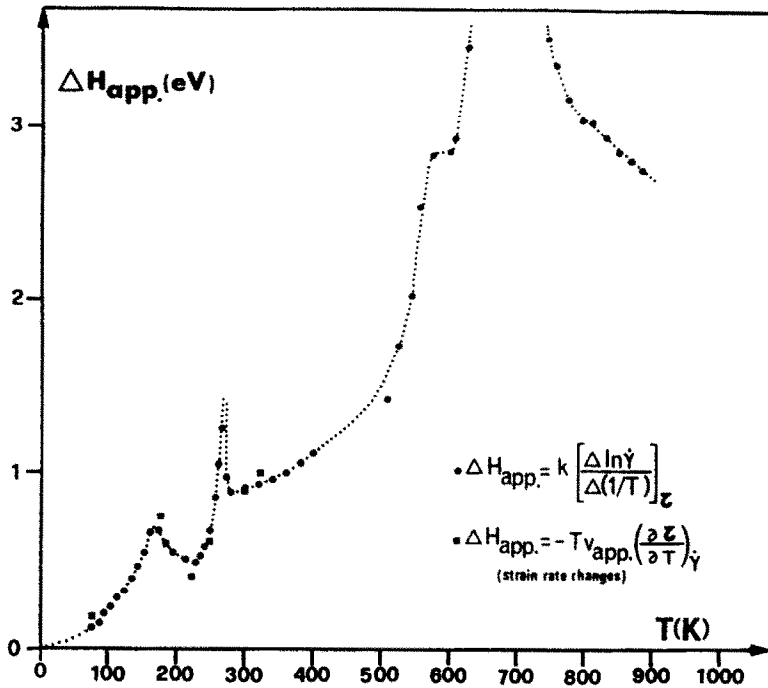


Fig. 5. The apparent activation enthalpy as a function of temperature.

Equations (3) and (4) suppose that the pre-exponential factor $\dot{\gamma}_0$ of the equation:

$$\dot{\gamma} = \dot{\gamma}_0 \exp(-\Delta G/kT) \quad (7)$$

is constant with temperature and stress. But Surek *et al.* [23, 24] and Heritier *et al.* [13] have shown that $\dot{\gamma}_0$ varies with temperature above about 600 K. That explains the bad agreement between results obtained from equations (3) and (4).

On the other hand, we can note that ΔH_{app}

increases from 0.1 eV (at 77 K) to 0.95 eV (at 300 K) and reaches 1.4 eV at 500 K. Same values have been found for Zr-O [3, 6, 8 & 10].

ΔG_{app} was computed using equations (5) and (6) and plotted against temperature in Fig. 6.

From equation (5), ΔG_{app} varies from 0.15 eV at 77 K to 0.6 eV at 300 K and reaches 1 eV at 525 K. From equation (6), ΔG_{app} varies between 0.15 eV at 77 K and 0.6 eV at 250 K. Therefore, equations (5) and (6) give approximately the same values for the

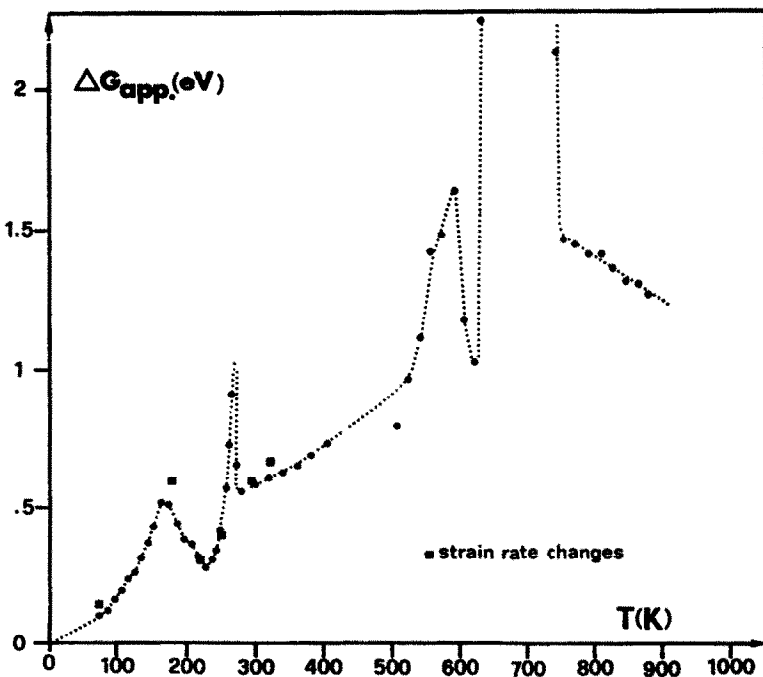


Fig. 6. The apparent activation free energy as a function of temperature.

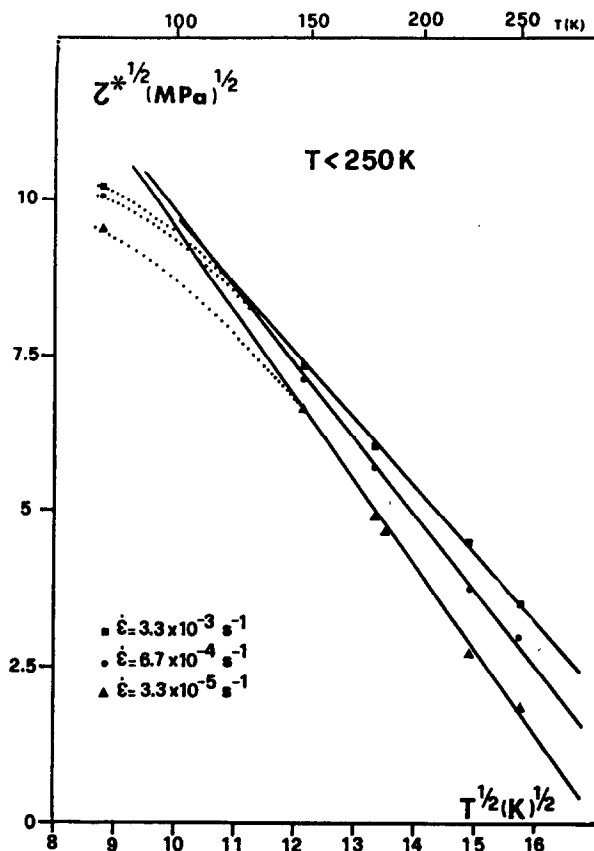


Fig. 7. A plot of $\tau^{*1/2}$ against $T^{1/2}$ (Fleisher's model) for the low temperatures range (77–250 K).

temperature range below 250 K. ΔG_{app} follows approximately the relation $\Delta G \sim 25 kT$ that is general for the metallic materials. We found again values similar to those of Zr–O [10] and Zr–H [7].

The two discontinuities on the ΔH_{app} and ΔG_{app} curves (between 180 and 250 K and between 600 and 700 K) (Figs. 5 and 6) show the transition between different mechanisms. In these transition zones, V_{app} , ΔH_{app} and ΔG_{app} cannot be defined because either the applied stress is athermal or several mechanisms take place.

4. DISCUSSION

4.1 Low and middle temperature range (77–600 K)

In the low and middle temperature range, it is clear from the flow stress temperature dependence [Figs. 2(A) and 2(B)] that two thermally activated mechanisms are operating and separated by a transition zone. Various flow control mechanisms were previously proposed for zirconium and its alloys [3–15]. For the concerned temperature range it is generally admitted that the plastic flow takes place essentially in the prismatic planes [25] and different propositions can be considered for the deformation mechanisms.

4.1.1 Low temperature range ($T < 180$ K). Two processes have been proposed:

1. Interactions between single interstitial solute atoms and dislocations [3, 5–8, 11, 12 & 26].
2. Peierls Nabarro mechanism [11].

Considering a solid solution strengthening mechanism, the Fleisher's model [27] could be used to describe the mechanical behaviour. In this model, the effective stress is given by the following relationship.

$$\tau^* = \tau_0^* (1 - (T/T_0)^{1/2})^2 \quad (8)$$

with $\tau^* = \tau - \tau_\mu$.

Where τ^* is the effective shear stress, τ_0^* the effective stress at 0 K, τ_μ † the athermal stress (equal to about 115 MPa in our case) and T_0 (250 K) is the critical temperature to which the first mechanism becomes 'athermal'. A plot of $\tau^{*1/2}$ versus $T^{1/2}$ (Fig. 7) indicates that equation (8) is not accurate at low temperatures. Furthermore for $T = 0$ K, the thermal stress τ_0^* has not the same value for all strain rates. Therefore, the Fleisher's model does not agree for describing the low temperature range behaviour.

Westlake [28] has shown that, if the dependence of obstacles spacing on the stress is taken into account [29, 30], the flow stress obeys the following relationship:

$$\tau^{*2/3} = A - B T^{1/3} \quad (9)$$

† The determination of τ_μ from data of Fig. 2(A) is difficult. However the choice of τ_μ has practically no effect on results of Fig. 7.

With

$$A = \left(\frac{N_z}{\mu b} \right)^{1/3} \cdot \frac{U_0}{4x_0 b} \quad (10)$$

and

$$B = A \left(\frac{3}{2} \cdot \frac{k}{U_0} \cdot \ln \frac{N b v_0}{N_z \dot{\gamma}} \right)^{2/3} \quad (11)$$

Where U_0 is the maximum activation energy required for a dislocation segment to overcome an obstacle, v_0 is the Debye frequency close to 10^{13} s^{-1} , x_0 the obstacle width, N the number of dislocations per unit volume held up by obstacles and N_z is the number obstacles per unit area in the glide plane (11 $\bar{2}$ 0); with 8100 at.ppm oxygen (1450 ppm by weight) the O interstitials density N_z was equal to $1.7 \cdot 10^{13} \text{ cm}^{-2}$.

Figure 8 shows a plot of $\tau^{*2/3}$ versus $T^{2/3}$ for different strain rates. There is a good agreement with equation (9), as in the work of Soo and Higgins [3]. For $T = 0\text{K}$, τ_0^* has sensibly the same value for all strain rates (i.e. $\dot{\epsilon} = 3.3 \times 10^{-5} \text{ s}^{-1}$, $\tau_0^* = 216 \text{ MPa}$; $\dot{\epsilon} = 6.7 \times 10^{-4} \text{ s}^{-1}$, $\tau_0^* = 224 \text{ MPa}$; $\dot{\epsilon} = 3.3 \times 10^{-3} \text{ s}^{-1}$, $\tau_0^* = 219 \text{ MPa}$). From the $\tau^{*2/3}$ vs $T^{2/3}$ curves the values of A and B have been obtained and substituted in equation (11). The slope of the low temperature range of ΔG_{app} vs temperature curve (Fig. 6) gives $(N b v_0 / N_z \dot{\gamma})$ and U_0 was found to be equal to about 0.77 eV. This value is in very good agreement with the

estimations obtained by Westlake for Zr-H-O (0.7 eV) and Soo and Higgins [3] for Zr-O (0.75 eV).

An estimation of x_0 can be obtained from equation (10):

$$x_0 = \frac{U_0}{4 b A} \cdot \left(\frac{N_z}{\mu b} \right)^{1/3}$$

With $\mu = 3.4 \cdot 10^{10} \text{ Pa}$, x_0 is of the order of $0.2b$. A value of $0.6b$ was reported by Tyson [9]. An estimation of the effective obstacle spacing can be made from the expression $\tau_0^* b l_0 x_0 = 0.77 \text{ eV}$ [12], a value of $80b$ or $28b$ is obtained corresponding to $x_0 = 0.2b$ and $0.6b$ and an activation volume $V_0 = x_0 l_0 b = 17b^3$ yields.

From the data of Fig. 4(A) and 4(B), it can be seen that the activation volume estimation is in very good agreement. If the oxygen atoms are randomly distributed l_0 is found to be equal to $15b$. Thus either the calculation of x_0 is invalid or a part of oxygen atoms is collected in clusters. Ruano *et al.* [6] has found that about 50% of oxygen atoms are in clusters. With this assumption $l_0 = 30b$. Another way to calculate l_0 is given by Friedel's formula (30):

$$l_0 = \left(\frac{2\mu_0 b}{N'_z \tau_0^*} \right)^{1/3} \quad (13)$$

with $N'_z = 8.5 \cdot 10^{12} \cdot \text{cm}^{-2} = N_z/2$

We found $l_0 = 34b$. It is probable that the effective obstacle spacing is equal to about $30b$ with an ob-

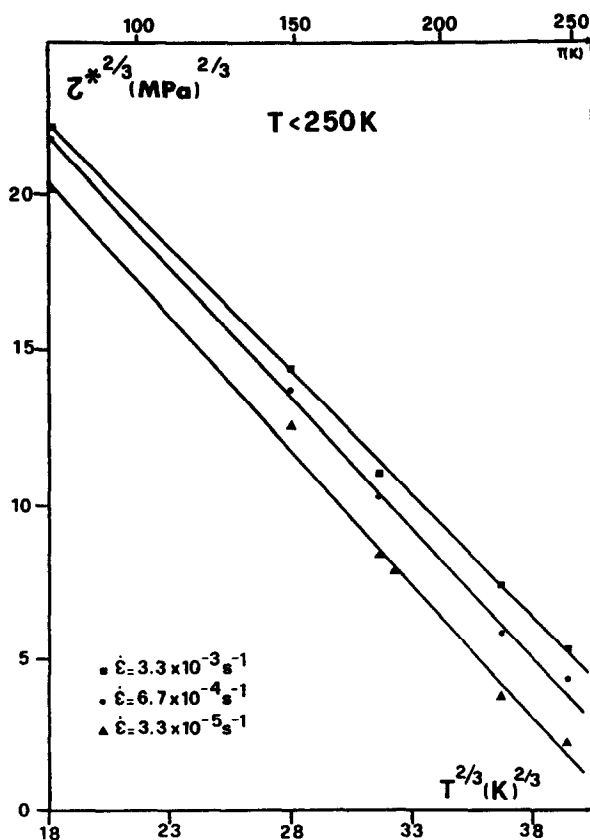


Fig. 8. A plot of $\tau^{*2/3}$ against $T^{2/3}$ (Westlake's equation) for the low temperatures range (77–250 K).

stacle width of $0.5b$ and that a part which is not negligible of oxygen atoms are clustered.

An estimation of the moving dislocation density ρ_m which is held up by dispersed obstacles can be obtained through the relation:

$$\dot{\gamma}_0 = N A b v = \dot{\gamma} \exp\left(\frac{\Delta G}{kT}\right) \quad (14)$$

where N is equal to ρ_m/l , $\dot{\gamma}_0 = 7 \cdot 10^5 \text{ s}^{-1}$ (this value is in good agreement with the one proposed by Conrad [34] $\sim 10^5 \text{ s}^{-1}$), A is the area swept by the dislocation following a successful activation event and the attempt frequency $v = v_0 b/l$ where v_0 is the Debye $A \sim N_z^{-1}$ ρ_m was obtained by substitution and given by (31):

$$\rho_m = \frac{\dot{\gamma}_0 l^2 N_z}{b^2 v_0} \quad (15)$$

ρ_m is in the order of 5.10^8 cm^{-2} , X-rays being used to measure the dislocation density after 2.5% cold work for zircaloy-4, a value of 10^{10} cm^{-2} was obtained [32]. The difference could be due to the use of other modes of deformation.

Therefore the law proposed by Westlake gives a very good description for the zircaloy-4 flow stress behaviour in the low temperature range. The flow is likely controlled by a solid solution strengthening mechanism. However, a Peierls Nabarro mechanism could be proposed as it can be seen above, the width of an obstacle is about $0.5b$ that seems in agreement with a Peierls barrier. This hypothesis would require to identify the effect of oxygen content on the Peierls Nabarro barrier. We can note that the apparent activation volume is lower than the one predicted for a Peierls Nabarro mechanism ($\approx 100 b^3$). In addition to our knowledge no experiments have yet been performed on zirconium of sufficient purity to show that the flow stress is not controlled by the interstitials content. Then a Peierls Nabarro process seems unlikely for this temperature range.

4.1.2. *Transition zone* (180–250 K). Considering that the first mechanism operating within the low temperature tends towards its critical temperature, the flow stress then becomes almost independent, typical of athermal regions and a second mechanism starts above about 250 K. Ruano *et al.* [6] have proposed that the controlling mechanism for the athermal domain could be one of the Orowan's stress type. A similar assumption could be adopted, where the stress required to overcome the clusters by the dislocation is given by [33]:

$$\tau_\mu \approx \frac{2\mu b}{L} \quad (16)$$

Taking $\tau_\mu = \tau_{\mu 250\text{K}} - \tau_{\mu 600\text{K}}$ ($\tau_{\mu 250\text{K}} = 115 \text{ MPa}$ and $\tau_{\mu 600\text{K}} = 37.5 \text{ MPa}$) $\mu = 3.4 \cdot 10^{10} \text{ Pa}$ and $b = 3.23 \text{ \AA}$, the calculated value of L (cluster spacing) is approximately equal to $870b$ which corresponds to a cluster density of $5 \cdot 10^{13} \text{ cm}^{-3}$.

4.1.3 *Middle temperature range* (250–600 K). Between 250 and 600 K, the different mechanisms which can control the deformation are the following:

1. Thermally activated overcoming of oxygen-atoms clusters and/or their dissolution with increasing temperature [6–15],
2. Lattice hardening [8],
3. The non conservative motion of jogs on screw dislocations [3],
4. Thermally activated cross slip [3],
5. Interactions between dislocation core and single interstitial atoms [5, 10, 12 & 14].

The last mechanism is unlikely because the resistance to flow offered by single interstitial atoms is negligible (above 250 K) [35]. If a cross slip mechanism operates above 250 K the theory requires a flow stress given by [3, 8]:

$$\frac{1}{\tau^* T} = A + \frac{B}{T} \quad (17)$$

Where A and B are constants. However the Fig. 9 shows that the flow stress behaviour does not obey equation (17), so that the cross slip mechanism is unlikely to control the deformation.

Soo and Higgins [3] have used a model proposed by Hirsch and Warrington [36]: above a critical temperature T_0 , the jogs advance with the aid of thermal activation. From this model $\Delta G'$, p , l and V can be estimated: $\Delta G'$ is the free energy of the defect created for every Burgers vector of advance of the dislocation and jog, p is the height of jog, l is the jog spacing and V the activation volume. We found that $\Delta G'$ is equal to about 4.1 eV, p to $5b$, l to $250b$ and V to $50b^3$. The activation volume value is in a good agreement with the data of Fig. 4(A) and 4(B). However the obtained $\Delta G'$ value is much higher than the one expected for a mechanism which operates at a temperature range below 600 K. Furthermore a thermally activated mechanism must operate from 0K and not from a finite T_0 . For all these reasons, this mechanism seems to be unlikely for controlling the deformation.

Considering Westlake's equation, a plot of $\tau^{2/3}$ versus $T^{2/3}$ shows again that this model gives a good description of flow behaviour (Fig. 10), U_0 is found to be 1.77 eV in this temperature range. This value is much more reasonable than 4.1 eV and the activation volume is of the same order (about $40b^3$) than that of the data in Figs. 4(A) and 4(B). In Ruano's work [6], τ is assumed to be dependent on $T^{2/3}$, that is in disagreement with the present work. This difference can be explained by considering that Ruano's specimens were not recrystallised (the microstructure was a bamboo structure characteristic of quenching).

Mills and Craig [10] have found that Fleisher's model described with a good approximation the flow stress behaviour. We have considered this model for the temperature range (250–600 K) but we have found a discord with this model. A Peierls Nabarro mechanism cannot operate in this temperature range as it was

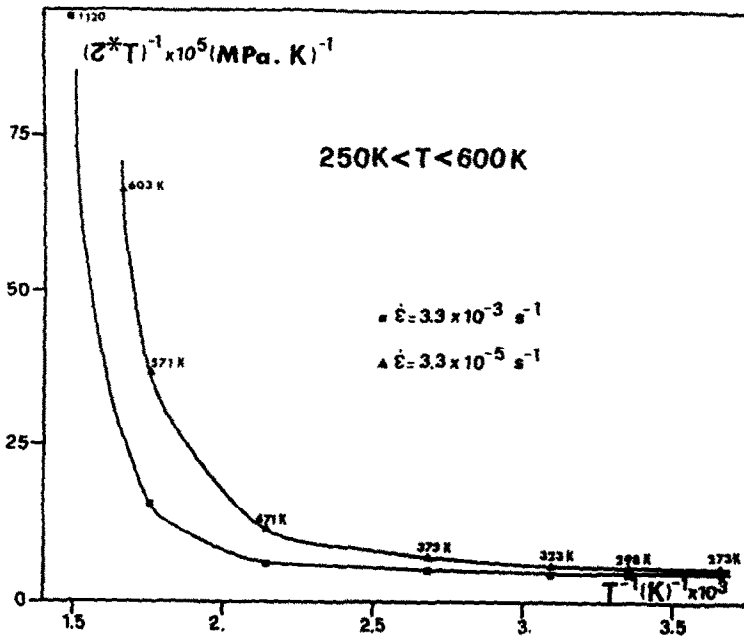


Fig. 9. A plot of $1/\tau^* T$ vs $1/T$ (cross slip mechanism) for the middle temperatures range (250–600 K).

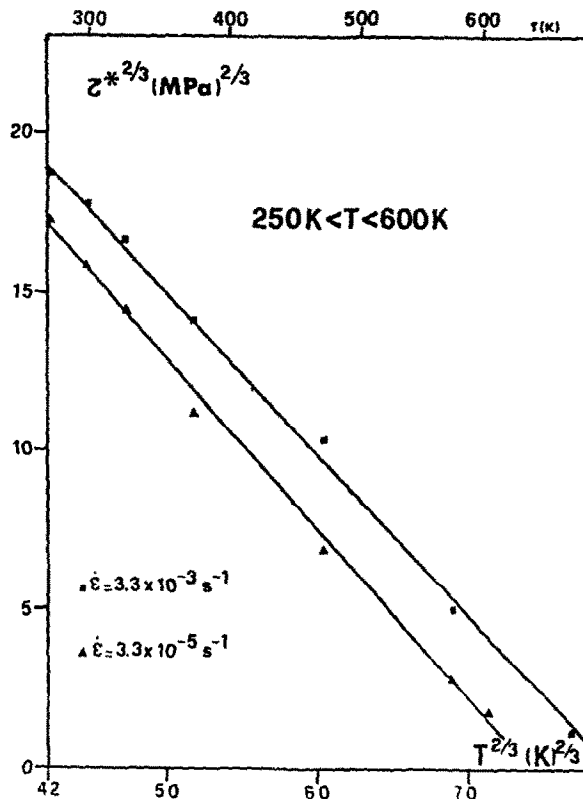


Fig. 10. A plot of $\tau^{*2/3}$ vs $T^{2/3}$ (Westlake's equation) for the middle temperatures range (250–600 K).

proposed by Das Gupta *et al.* [8] for the same reasons exposed paragraph 4.1.1.

Therefore, for the temperature range (250–600 K) the strain rate controlling mechanism is in a very agreement with Westlake's model. This supposes that the flow is controlled by interactions between disloca-

tions and clusters of several interstitial atoms since a single atom does not interact with a dislocation at these temperatures.

The second mechanism and the 'athermal' one are assumed to be parallel [53] (this means that the mechanism which controls the deformation requires

the lower stress). The 'athermal' process critical temperature is T_{02} . The first mechanism and the athermal one are assumed to be consecutive (this means that the process which controls the deformation requires the higher stress). The first mechanism critical temperature is T_{01} . As T_{01} and T_{02} are close together we cannot clearly observe the athermal plateau.

4.2 Athermal region (600–700 K)

The transition from the preceding thermal rate controlling mechanism to an athermal region appears to occur between 600 and 700 K. The athermal stress component can likely be attributed to the dislocation unpinning on the dislocations network nodes by an Orowan or a Frank and Read's mechanism [48]. An inverse strain rate effect appears in this region. A similar behaviour was found by Heritier *et al.* for Zr–O [13], by Thorpe *et al.* for Zr–Nb [15] and by Tyson *et al.* for Zr–N [12]. The increase of flow stress versus temperature was attributed by these authors to dynamic strain ageing.

Indeed, many investigators, studying the internal friction behaviour in zirconium alloys [39–45], have shown the existence of several damping peaks in the same temperature range. These peaks are assumed to be due to a Snoek ordering of substitutional-interstitial solute atoms pairs in the applied stress field. The interstitial atoms can be nitrogen or oxygen [43] which give peaks between 600 and 700 K. As the location of the peak of the bell plot is a function of strain rate and temperature, a thermally activated mechanism can be suggested and the activation energy ΔH (if the temperature of maximum is related to the corresponding strain rate through an Arrhenius term) is found to be about 55 kcal/mole (2.39 eV). Indeed Heritier *et al.* [13] and Thorpe *et al.* [15] have found 51.4 (2.23 eV) and 51 kcal/mole (2.21 eV) respectively. From the literature [46, 47], the lattice diffusion activation energy of oxygen is assumed to be in the range 44–55 Kcal/mole (1.9–2.4 eV). As in zircaloy-4, the oxygen content is much greater than the nitrogen one (8100 ppm against 160 ppm) it can then be reasonable to suppose that the strengthening is due to Snoek's ordering of S–i or i–i oxygen atoms pairs.

4.3 The temperature range above 700 K

In this temperature range, the flow stress decreases in a sharp way as the temperature is increased. Many mechanisms have been proposed to describe the flow stress behaviour. Huang *et al.* [49] who has studied load-relaxation of zircaloy-4 has proposed that grain boundary sliding controlled the deformation from 700 K. Surek *et al.* [23] suggests that the rate controlling process in the high temperature was the unpinning of dislocations at the dislocations network nodes. Tyson and Craig's work showed that the flow stress obeyed equation (17), with a good agreement; that would mean that cross slip operates in this temperature range. Some observations of fracture mode of zirconium alloys have shown that, from about

700 K, the specimen failure appeared to be made by voids formation [50, 51]. That suggests creation and coalescence of vacancies to form cracks. We know that zirconium shows an abnormal high density of vacancies, at the transus temperature (1100 K) about 10^{-4} [52]. This means that the creation of vacancy is very easy (by non conservation motion of jogs for example). The last hypothesis is therefore, realistic. Furthermore the Frank's dislocations network could likely evolve by recovery.

In summary, it is difficult to choose between these mechanisms. We do not have at our disposal enough data to choose reasonably. More complete experiments must be made. However, it can be remarked that several mechanisms could be operating, otherwise no V , ΔH and ΔG could be defined.

5. CONCLUSIONS

After tensile stress tests at temperature range 77–900 K and various strain rate ($3.3 \cdot 10^{-5} \text{ s}^{-1}$, $6.7 \cdot 10^{-4} \text{ s}^{-1}$ and $3.3 \times 10^{-3} \text{ s}^{-1}$) 5 zones could be distinguished on the flow stress curves:

1. The first zone is included between 77 and 180 K where the rate controlling mechanism is likely the overcoming of interstitial atoms by the moving dislocations with an activation energy $U_0 = 0.77 \text{ eV}$.
2. Between 180 and 250 K the first mechanism tends towards its critical temperature and the athermal flow stress component is due to the overcoming of clusters of interstitials atoms by an Orowan's mechanism.
3. For the temperature range 250–600 K the deformation can be controlled by the thermally activated overcoming of interstitial atoms cluster and/or their dissolution with increasing of temperature.
4. From 600–700 K the stress is observed to become insensitive to temperature then it increases with temperature increasing. The athermal stress component is most likely to be associated to an Orowan's type mechanism which is required to bow out segments of the Frank's network. The strengthening mechanism is attributed to a Snoek's ordering of oxygen atoms.
5. At high temperature (above 700 K), the thermal flow can be controlled by one or several mechanisms which are parallel to the last one (unpinning of dislocations at the nodes, recovery evolution of the Frank's network, vacancies coalescence to form cracks, cross slip and/or grain boundary sliding).

Within the studied range of temperature it can be concluded that the mechanical behaviours of zirconium–oxygen alloys and zircaloy-4 are very similar.

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REFERENCES

1. Garde A. M., Ghung H. M. and Kassnev T. F. *Acta Metall.* **26**, 153 (1977).
2. Kearns J. J., McCanley J. E. and Nichols F. A. *J. nucl. Mater.* **61**, 169 (1976).
3. Soo P. and Higgins G. T. *Acta Metall.* **16**, 177 (1968).
4. Ramachandran V. and Reed-Hill R. E. *Metall. Trans.* **1**, 2105 (1970).
5. Tiwari S. N., Lloyd D. J. and Tangri K. *Metall. Trans.* **3**, 2605 (1972).
6. Ruano O. and Ellsner G. *J. less common Metals* **40**, 121 (1975).
7. Soo P. and Higgins G. T. *Acta Metall.* **16**, 187 (1968).
8. Das Gupta P. and Arunachalam V. S. *J. Mater. Sci.* **3**, 271 (1968).
9. Tyson W. R. *Can. Metall. Q.* **6**, 301 (1967).
10. Mills D. and Craig G. B. *Trans. Met. Soc. AIME* **242**, 1881 (1968).
11. Sastry D. H., Luton M. J. and Jonas J. J. *Phys. stat. sol.* (a) **33**, 173 (1976).
12. Tyson W. R. and Craig G. B. *Can. Metall. Q.* **7**, 119 (1968).
13. Heritier B., Luton M. J. and Jonas J. J. *Metall. Sci.* **8**, 41 (1974).
14. Lee D. *Metall. Trans.* **1**, 1607 (1970).
15. Thorpe W. R. and Smith I. O. *J. nucl. Mater.* **78**, 49 (1978).
16. Schwoppe A. D. and Chubb W. *Mechanical properties of iodide zirconium alloys*, Report BMI-780, (1952).
17. Luton M. J. and Jonas J. J. *Can. Metall. Q.* **11**, 79 (1972).
18. Veevers K. *J. nucl. Mater.* **55**, 109 (1975).
19. Rheem K. S. and Park W. K. *J. Korean nucl. Soc.* **8**, 19 (1976).
20. Luton M. J. and Jonas J. J. *Proceedings Second International Conference on the Strength of Metals and Alloys. Asimolor, California*, 30 Aug-4 Sept., Vol. 3, Am. Soc. Metall. p. 1100 (1970).
21. Evans A. G. and Rawlings R. D. *Phys. stat. sol.* **34**, 9 (1969).
22. Schoeck G. *Phys. stat. sol.* **8**, 499 (1965).
23. Surek T., Luton M. J. and Jonas J. J. *Phys. stat. sol.* (b) **57**, 647 (1973).
24. Surek T., Kuon L. G., Luton M. J. and Jonas J. J., J. E. Dorn Memorial Symposium *Rate Processes in plastic deformation*, Oct. 16 and 17 (1972), Cleveland, Ohio.
25. Rapperport E. J. and Hartley C. S. *Trans. Met. Soc. AIME*, **218**, 869 (1960).
26. Tyson W. *J. nucl. Mater.* **24**, 101 (1967).
27. Fleischer R. L. *Acta Metall.* **15**, 1513 (1967).
28. Westlake D. G. *Acta Metall.* **12**, 1373 (1964).
29. Seeger A. *Phil. Mag.* **40**, 1194 (1955).
30. Friedel J. *Dislocations* p. 224, Addison-Wesley, New York (1964).
31. Hirth J. P. and Nix W. D. *Phys. stat. sol.* **35**, 177 (1969).
32. Aqua I. N. and Owens C. M. *Trans. TMS. AIME* **239**, 155 (1967).
33. Friedel J. *Dislocation* p. 372, Addison-Wesley, New York, (1964).
34. Conrad H. *High Strength Materials* (edited by F. Zackay) p. 436, Wiley, New York (1965).
35. Tyson W. R. *Can. Metall. Q.* **6**, 301 (1967).
36. Hirsch P. B. and Warrington D. H. *Phil. Mag.* **6**, 735 (1961).
37. Dorn J. E. and Rajnak S. *Trans. Metall. Soc. AIME* **230**, 1052 (1964).
38. Christian J. W. and Masters B. C. *Proc. R. Soc. A* **281**, 240 (1964).
39. De Paula e Silva E., Com. Nougue J., Beranger G. and Lacombe P. *Scripta Met.* **5**, 795 (1971).
40. Bedford A. J., Fuller P. G. and Miller D. R. *J. nucl. Mater.* **43**, 164 (1972).
41. Cacougnolle J. L., Sarrazin S. and de Fouquet J. *J. Phys.* **7**, C2, 21 (1971).
42. Browne K. M. *Acta Metall.* **20**, 507 (1962).
43. Mishra S. and Asundi M. K. *Can. Metall. Q.* **11**, 69 (1972).
44. Fernandez L. and Povofo F. *J. nucl. Mater.* **66**, 79 (1977).
45. Ritchie I. G., Rosinger H. E. and Atrons A. *J. nucl. Mater.* **62**, 1 (1976).
46. Ritchie I. G. and Atrons A. *J. nucl. Mater.* **67**, 254 (1977).
47. Pemsler J. P. *J. elec. Soc.* **105**, 315 (1958).
48. Luton M. J. and Jonas J. J. *High temperature deformation of zirconium and zirconium tin alloys*. Dep. Met. Eng. McGill University.
49. Huang F. H., Sabol G. P., McDonald S. G. and Li C. Y. *J. nucl. Mater.* **79**, 214 (1979).
50. Sabol G. P. and McDonald S. G. *J. nucl. Mater.* **64**, 223 (1977).
51. Lee D., Knoch E. F. and Rogers H. C. *Metall. Trans.* **1**, 3125 (1970).
52. Carpenter G. J. C. and Watters J. E. *Acta Metall.* **21**, 1207 (1973).
53. Li J. C. M. *Rate processes in plastic deformation of materials No. 4 Am. Soc. Metall.* pp. 479-496 (edited by Li J. C. M. and Mukherjee A. K. (1975)).